The Rise of Synthetic Visual and Auditory Data: A Multimodal Framework for Synthetic Media Detection

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ABSTRACT

With the rise of generative AI tools like ElevenLabs, a popular text-to-speech software capable of cloning and generating voices and VEO 3, Google's state of the art video generation tool with native audio generation, combined with Deepfakes, poses a huge risk towards Cybersecurity, specifically ethical and security risks. These technologies are evolving and progressing at an alarming rate, and as the world continues to be ever more interconnected with one another, heavily relying on digital communication, including the Philippines, the line between authentic and synthetic media is rapidly blurring. This study is a quantitative experimental design proposing a model for a multimodal deep learning detection system for Sora 2 and Veo 3.

Chapter I

INTRODUCTION

In today's era, we live in a world that is interconnected and digitally dependent and we rely on technology to make the modern world run as we know it. Technology is in every facet of our modern life, from communication and education to businesses and governance, as well as our personal identity with social media applications. This era that we are in is called the Information Age, more commonly known as the Digital Age and it is considered as the "Third Industrial Revolution" [1]. The rise of the internet, cloud computing, and digital systems have reshaped the way we live our lives. However, for an era that is called the "Information Age", that very foundation, information, is being called into question and faces an imminent threat that has been looming for decades.

As generative AI technologies continue to progress, we are entering a phase in our era, where the authenticity of digital content can no longer be taken for granted. That is not to say that user generated content and forgery has never existed during the pre-internet age or during its infancy, but generating voices, images, videos, and even entire conversations can now be artificially forged/generated with accuracy never seen before. Generative AI tools like ElevenLabs, a very popular Text-To-Speech software that uses machine learning to clone someone's voice and even generate a convincing voice of a person that doesn't exist [2] and VEO 3, Google's state of the art video generation tool with audio generation [3]. If combined with Deepfakes, a hyper-realistic video digitally manipulated to mimic a certain person, is also evolving at an alarming speed [4] and poses a huge risk towards Cybersecurity, specifically

ethical and security risks. We are approaching a new era, perhaps an era that is even more destructive and transformative than the last.

The biggest threat we face now is how it can be used in social engineering: creating realistic clones of voices, faces and conversations which can easily fool any humans, making it nearly impossible to distinguish fakes from reality. This technology is already here, but it will be in its worst state, as technology always rapidly improves and doesn't remain stagnant as is. This study will explore the implications of these tools on digital authenticity and their impact on a digitally dependent society.

Gap / Opportunity

In 2023, there was an average of 1,530 percent increase in deepfake incidents in the Asia-Pacific region compared to the previous year, with the Philippines experiencing the most significant cases globally at 4,500 percent [4]. A more recent deepfake controversy in the Philippines, Senator Ronald "Bato" Dela Rosa, a confidant/ally of Rodrigo Duterte, shared an AI-generated video on June 11, 2025. This is from a report by Sturt [5], which depicted two male Filipino students expressing strong support for Vice President Sara Duterte, who is currently facing an impeachment trial. The video depicts that the charges against Duterte were "politically motivated" and that:

“politicians who are serious about rooting out corruption should look at themselves first.” [5]

With the recent release of Sora 2, September 2025, the successor of OpenAI's initial Sora version which was released back in December, 2024 [6]. It now has the capability to generate both visual and auditory data from just a few snippets of your voice recording and a face scan of your face. Their latest video generation model is more physically accurate, realistic, and more controllable than prior systems with synchronized dialogue and sound effects [7]. The generated videos do come with a watermark, it is visible on the entire video that was generated, but many different programs can erase that visible watermark very easily. This leads to videos that are AI generated, without any visible watermark, and are intentionally created and tampered for malicious use.

The clock has started to tick, just recently as this review paper was getting written, an article report [8] published on July 18th, 2025. An elderly woman in Florida, received a call from an unidentified number detailing that her daughter had killed someone while texting and driving. She was conned out of $15,000 (₱ 851,000) by a scammer who used artificial intelligence to replicate her daughter's voice. The elderly woman stated:

"She had heard her daughter's hysterical voice claiming she had hit a pregnant woman with her car while texting and driving. "There is nobody that could convince me that it was not [her voice], I know my daughter's cry, even though she's an adult, I still know my daughter's cry."" [8]

In the time where the information at hand may or may not be real, it's always safe to remain cautious. The article [8] suggests that we should make "Safe Phrases" to guard against voice cloning. It was further claimed that:

"It only takes three seconds of audio content for an AI to clone someone's voice. And one call from a scammer to trick you into thinking a 'friend' needs money." [8]

This is a reliable and known method for protecting against scams, but you need to verify it now with voice, if they ever ask for something out of the blue.

We are now approaching an era where information online may not appear to be as what you see. Generative AI is getting more proficient at creating videos that can sound and look realistic in a matter of seconds. The history of generative AI is very recent; chatbots were the first application of the technology in the 1960s. Currently, this type of artificial intelligence can create synthetic data, videos, audio, images, and text of excellent quality in a matter of seconds. This paper begins by addressing the implications of generative AI on digital authenticity and how we can better be guarded against such tools in detecting telltale signs and artifacts of AI generated audio and videos.

Statement of the Problem

The proliferation of high-quality/fidelity AI generated images and videos, makes it harder to detect and be able to distinguish the real videos from the fakes. With the rise of generative AI tools like ElevenLabs, a popular text-to-speech software capable of cloning and generating voices and VEO 3, Google's state of the art video generation tool with native audio generation, combined with Deepfakes, poses a huge risk towards Cybersecurity, specifically ethical and security risks. With the release of Sora 2, these technologies are evolving and progressing at an alarming rate, and as the world continues to be ever more interconnected with one another,

heavily relying on digital communication, including the Philippines, the line between authentic and synthetic media is rapidly blurring. Current systems used in detection are very specialized, and only work well with audio or visual analysis. Current systems also have trouble staying accurate when they come into new generative approaches or real-world situations like video compression.

This study tackles the problem of improving current detection systems, and to accommodate the rapid development of generative AI tools of today's age. The technical challenge is to improve upon a model that can effectively combine/fuse the features from both visual and auditory generated content to make a more accurate and up-to-date classification of these newly released tools into the public's hands. The end result would not be merely to detect fakes, but to develop an enhanced methodology that can classify and detect both audio and video/image from existing single-modality approaches, particularly on compressed and state-of-the-art synthetic/generated media, that has been plaguing social media.

Specifically, it aims to answer the following questions:

1) What deep learning method is the best suited to detect tampered AI generated videos from Google, ElevenLabs and OpenAI and their respective model weights and parameters.
2) What hybrid deep learning model should be used in detecting both visual and auditory generated data to classify telltale signs of AI-generated content.
3) How will we be able to classify and separate real videos from AI-generated content and how accurate it would be compared to previous/pre-existing techniques.

Objectives

We live in a world that is interconnected and digitally dependent and we rely on technology to make the modern world run as we know it. The world has become accustomed to digital life, everything runs in the cloud, on the internet, in the systems we use for everyday life. As generative AI technologies continue to progress, the tools that can be leveraged against us are also progressing. There are no major incidents yet in the Philippines pertaining to the use of Generative AI and Deepfakes towards cybersecurity, but it is only a matter of time until one falls for such tricks, especially considering the rate of progress we've seen in recent times. No one is infallible, one day or sooner, we will eventually fall for such trickery, and the only thing we can do now is to familiarize ourselves with the common telltales of AI generated videos and image artifacts.

Sora 2 is the newly released product from OpenAI, and we saw the sudden strong interest in generative AI today from businesses, consumers and industry players alike. With billions of dollars in new investment, the pace of progress and innovation accelerated dramatically and we are running straight into a future no one is certain of. A popular line from the movie Jurassic Park, perfectly describes the situation we are in. The line goes as "Yeah, yeah, but your scientists were so preoccupied with whether or not they could that they didn't stop to think if they should." This sentiment perfectly encapsulates what came after text generation, it was the eventual creation of image, audio and video generation capabilities. The primary objective of this study is to design/improve, implement, and evaluate a multimodal deep learning model for the detection of both AI-generated video and audio content.

Specifically, this study aims to:

1) To improve and implement a hybrid deep learning architecture that combines Convolutional Neural Network (CNN) for spatial feature extraction from video frames with either LSTM/RNNs (CNN-RNN/LSTM) or a Transformer-based network for analyzing temporal features in the audio spectrogram.
2) To use both synthetic auditory and video data from newly released tools like Sora 2, VEO 3 and authentic data from real videos and audios.
3) To evaluate the proposed hybrid model and rigorously evaluate its performance against a baseline video-only detector using standard metrics (Spatial and Temporal consistency, motion quality, F1-Score, etc…)

Significance of the Study

The escalating threat of high quality/fidelity generative AI tools that are getting made and being publicly released should be a cause of concern for many. That doesn't seem to be the case for many youths, some people see it as bad whilst some people see it as good for the public, as it democratizes these tools for everyone to use and to get to know about what creativity can they unleash. Many don't have the artistic skills to showcase what passions/imaginations they have, and these tools give them a medium for that use-case. What they fail to realize is the ramifications of such generative AI tools, especially when bad actors get their hands on these publicly and readily available tools, that could generate synthetic data in seconds, and especially in a world that is digitally dependent, and filled with misinformation online. By being able to classify and differentiate what is authentic from synthetic, the results provides significance in the following areas:

1) Technical Contribution to CS Domains – It will contribute to the current detection methodology available today, with a more up-to-date dataset such as OpenAI's newly released Sora 2. It will analyze both visual and auditory data, for a more accurate classification if it is synthetic or authentic.
2) Applications in Cybersecurity – The findings and the developed model will be highly valuable for cybersecurity professionals, and it can be potentially improved upon and used in a more stable and professionally built framework.
3) New Academic Knowledge – This research will generate empirical data on the effectiveness of detecting multi-modal detection against the latest generation of AI models. Especially current generation tools that are able to generate both audio and visual data on the fly, technology which was not possible beforehand.

Scope and Limitation of the Study

The scope of this study paper is centered on the improvement and evaluation of a multimodal hybrid deep/machine learning framework for the detection of AI-generated synthetic content. The study paper will primarily use publicly available video datasets, and synthetic video and audio content will be generated using OpenAI's Sora 2 and or Google's Veo 3, as user-generated content is publicly available and accessible to everyone.

The technique will process both visual and auditory inputs from videos, using machine learning techniques like Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs) for audio synthetic data, it can also use a Transformer-based model for turning

audio into spectrograms for analyzation. The extracted features from techniques used will be evaluated and classified by the improved framework to distinguish and differentiate between authentic and AI-generated user content.

The scope of this research is limited to the technical development/advancement of currently available generative AI tools today, and the dataset is exclusively made for VEO 3 and Sora 2. Furthermore, the research will be confined to visual data and it will not include textual synthetic data. It is only framework-level, it is by no means production-ready.

Chapter 2

REVIEW OF RELATED LITERATURE

Artificial Intelligence, or more specifically, Generative AI is rapidly evolving. The first public version of ChatGPT by OpenAI came out in 2022, it has been a little over 3 years, and it has progressed at an astonishing rate. The same can be said in synthetic image and video generative tools.

Table 1. Rate of Progress of Generative AI Image and Video Tools

Model	Year	Developer	Development Time
DALL-E	2021	OpenAI	Baseline
DALL-E 2	2022	OpenAI	~15 months (Jan 2021 - Apr 2022)
Imagen	2022	Google	~4 months (Apr 2022 - Aug 2022)
Stable Diffusion	2022	Stability AI	~4 months (Aug 2022 - Aug 2022)

Midjourney V5	2023	Midjourney	~7 months (Aug 2022 - Mar 2023)
Runway Gen-2	2023	RunwayML	~3 months (Mar 2023 - Jun 2023)
Veo 2	2024	Google DeepMind	~11 months (Jun 2023 - May 2024)
Sora	2024	OpenAI	~0 months (concurrent with Veo 2)
Imagen 3	2024	Google DeepMind	~0 months (concurrent with Veo 2, Sora)
Veo 3	2025	Google DeepMind	~5 months (Dec 2024 - May 2025)
Sora 2	2025	OpenAI	~4 months (May 2025 - Sep 2025)

This paper is all about synthetic media detection, which is a subset of Multimedia forensics, all under Digital Media forensic and Computer Vision. The current status of the field is an ongoing "arms race" or the cat and mouse game for the lack of a better term, between increasingly sophisticated generative/diffusion models (like Sora 2, VEO 3) and the detection models designed to counter them [9]. As generative models become more realistic and accurate, detection methods must evolve from identifying simple visual flaws to understanding the subtle physical, temporal, and spatial inconsistencies, most of which are present from current synthetic media today.

Deepfakes have been around ever since the creation of the internet, as mentioned by Regan [10] *"The concept of deepfakes (or deepfaking) can be traced back to efforts starting in the 1990s, when researchers used CGI in attempts to create realistic images of humans."* This technology has gained more traction during the 2010's, since we saw a considerable amount of

large datasets readily available online / on the internet. We've also seen the advancement with machine learning, which powered a whole new way of computing resources which subsequently led to major advancement in the field. Regan also asserted that the point of no return is attributed to the 2014 breakthrough in deep learning, with the new machine learning concept called Generative Adversarial Network (GAN) which supercharged deepfakes even more. What truly spurred the development for the current, transformative era of generative AI was the 2017 paper from Google researchers, titled "Attention Is All You Need" [11]. This paper introduced the Transformer architecture which all modern video and image generation models use. Before the Transformer, RNNs processed data sequentially (one at a time), which made it slow. However, the Transformer architecture allowed a model to weigh the importance of all parts of the input data simultaneously, which allowed for massive parallelization in training and unlocked the scaling that led directly to large language models like GPT, and subsequently, to the image and video generation models like DALL-E and Sora [12]. GANs lit the fire, but the Transformer architecture made the entire modern generative AI ecosystem not only walk, but to sprint at the finish line.

With the recent release of Sora 2, developed by OpenAI, It now has the capability to generate both visual and auditory data from just a few snippets of your voice recording and a face scan of your face. Their latest video generation model is more physically accurate, realistic, and more controllable than prior systems with synchronized dialogue and sound effects [7]. Many conferences and journals are hosted on CVPR (Conference on Computer Vision and Pattern Recognition), ICCV (International Conference on Computer Vision), and NeurIPS (Conference on Neural Information Processing Systems); most published works can be found on

ArXiv. The research challenges are the rate of progression that these tools are undergoing, as well as the active malicious tampering of data sourced from the Internet, which can tamper with the training data during data collection processes, as well as video compression.

Terminology

This section explains the different terminologies and key terms used in AI-generated video detection, and core concepts of the underlying technologies that power generative image and video tools.

1) Generative AI: New class of AI models capable of creating new, original content based on patterns learned from its training data (like images, text, audio, video) [13].
2) Deepfake: A type of synthetic media where a person's own likeliness is replaced/mimicked with another person. The term is now often used broadly/associated to refer to any AI-generated video or audio [14].
3) Generative Adversarial Networks (GANs): It is a deep learning architecture. It trains two neural networks to compete against each other to generate more authentic new data from a given training dataset [15].
4) Deep Learning: It is an artificial intelligence (AI) method that teaches computers to process data in a way inspired by the human brain [15].
5) Diffusion Model: Newer class of generative models, rapidly gaining traction for their exceptional performance in generating high-quality outputs. Rather than competing adversarially, like GANs, they progressively transform simple noise into structured, detailed data, mimicking real-world patterns. [16].

6) Convolutional Neural Network (CNN): A type of deep learning model that is specialized for image and video processing. It's the standard for extracting features from images or video frames [17].
7) Recurrent Neural Network (RNN) / LSTM: Designed for sequential data with recurrent connections that allow information to persist across time steps. It is particularly useful for modeling time dependencies [17].
8) Transformer: A more modern architecture, a type of neural network architecture that excels at processing sequential data and uses attention mechanisms to weigh the importance of different parts of the input. It has shown great promise in both generation and detection and in other fields of artificial intelligence (AI), such as computer vision, speech recognition [18].
9) Few-Shot Learning: A machine learning framework in which an AI model learns to make accurate predictions by training on a very small number of labeled examples [29].
10) Multimodal Detection: A detection approach that analyzes multiple data streams (visual frames and the audio) simultaneously to attain accurate classification / percentage score.

General Research Framework

This section explains the General process of a multimodal approach on developing synthetic media detection. The work done in [27] follows a multimodal framework pipeline but focused primarily on Deepfake materials and not AI generated media.

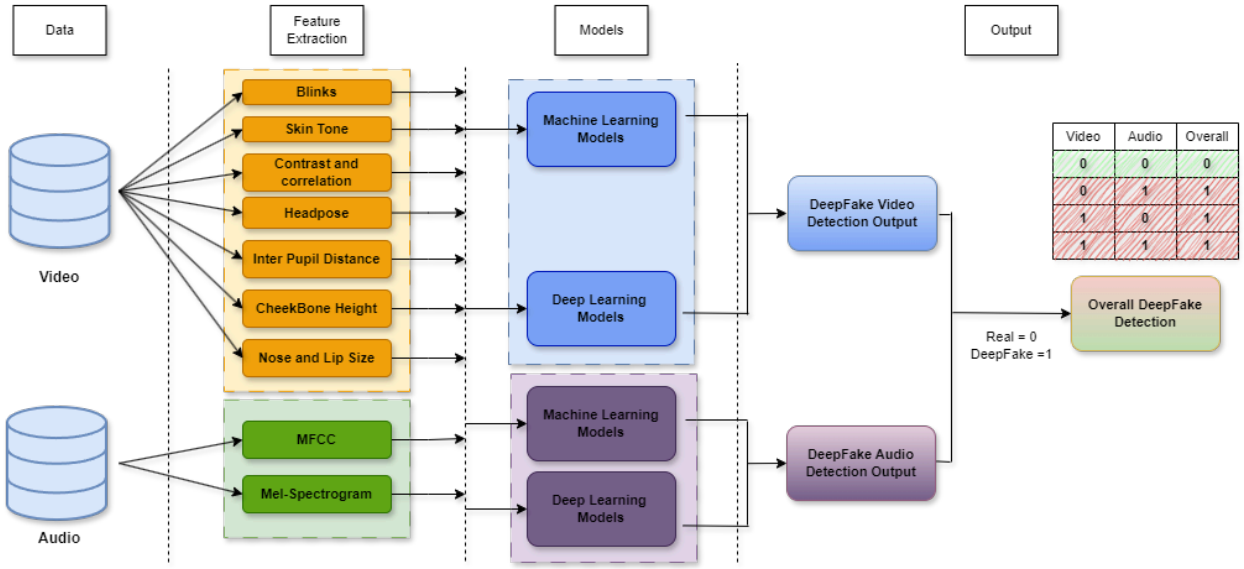


Figure 1. Pipeline for proposed framework shown in [27].

1) Data Collection: Gathering a large dataset of both authentic and synthetic media. It will involve collecting real world videos and generating a corresponding set of fakes mainly using Sora 2. Usually, most AI video generators can start with an image as a baseline for what video is getting generated.
2) Pre-processing: Standardizing the collected data into the following steps;
    a) Frame Extraction: Slicing the videos into individual image frames.
    b) Audio Extraction: Separating the audio track from the video.
    c) Normalization: Resizing frames and normalizing pixel values.
    d) Feature Conversion: Converting audio waveforms into spectrograms.
3) Feature extraction: Using a deep learning model to automatically learn the discriminative features that separate authentic from synthetic data. Will primarily focus on the noise of background texture and not physical traits. CNNs are widely used on frames/spectrograms.

4) Models to be used: We will need to design and train a classifier. This can be either a simple classifier on top of a CNN or a more complex model (like an LSTM or Transformer) that analyzes the sequence of extracted features.
5) Testing and Evaluation: Assessing the trained model's performance on a different, unseen test dataset using standard metrics like precision, recall, F1 score, and accuracy [27].

Survey Paper

No.	Ref	Scope	Dataset	Methodology	Result	Further Research
1	[19]	Assesses 3D visual coherence in synthetic videos, focuses on appearance, motion, and geometry.	Paired real/synthetic sets (160k videos), benchmark with Sora/Kling videos	L3DE uses 3D ConvNet with proxies (DINOv2, RAFT, UniDepth)	Fusion score correlates 0.7566 with reconstruction quality; 70-83% deepfake detection accuracy.	Expand datasets, handle longer videos, integrate with data-centric approaches.
2	[20]	Detects temporal artifacts across generators including Sora.	GVF (964 triples, real-fake pairs from 9 models including Sora)	DeCoF uses CLIP features with transformers for consistency verification	96.87% accuracy on Sora.	Extend to Sora 2, longer videos, ethical misuse prevention.
3	[21]	Multi-label classification of artifacts in Sora videos	300 annotated frames from 15 Sora videos.	2D CNNs (ResNet-50, EfficientNet) for multi-label classification; Grad-CAM for visualization.	94.14% accuracy; object mismatches most common (80.3%).	Larger datasets, temporal analysis (3D CNNs), limited-data detection.
4	[22]	Detection of diffusion-generated videos with focus on temporal dynamics.	6k+ videos from SVD, Sora etc... real from MiT/Video-ACID.	DIVID uses DIRE + RGB with CNN-LSTM; two-phase training.	93.7% in-domain accuracy; up to 16% improvement out-of-domain.	Optimize timesteps, adversarial robustness, dataset release.

5	[23]	Comprehensive review of detection methods, including videos, transitioning from specialized to MLLM-based.	References multiple (GenVideo, AV-Deepfake1M).	Taxonomy of Non-MLLM (CNN/LSTM) and MLLM (CoT, watermarking) approaches.	MLLMs ~80% accuracy with explainability; challenges in video localization.	Hybrid methods, broader modalities, ethical frameworks.
6	[24]	Benchmark for cross-generator detection, state of the art AI video generator models.	143k videos from 8 generators; semantic labels.	Binary classification with CNN/Transformers.	79.9% accuracy; drops to 50-60% cross-generator.	Scenario-specific detectors, attribute impact analysis.
7	[25]	Shows image detectors fail on videos; learns video-specific traces.	6k+ videos from Luma, SVD, Sora, etc…	CNNs (MISLnet) with patch aggregation.	0.983 AUC detection; 0.991 source attribution.	Real-world integration, more generators.
8	[26]	A general two-stream network for detecting both spatial and temporal artifacts.	FF++, Celeb-DF	Two-stream CNN: one for frame appearance, one for optical flow (motion). MesoNet (CNN variants)	Showed that combining motion and appearance improves detection accuracy. Up to 98.4% accuracy.	Explore more efficient temporal modeling beyond optical flow.
9	[27]	Multimodal detection fusing audio, video, and temporal streams.	DeepFake-TIMIT, FF++	Hybrid DL with separate streams for audio and video, fused with a temporal network.	state-of-the-art results by leveraging information from both modalities. ~80-95% AUC benchmark	Improve the efficiency of the fusion mechanism.
10	[28]	Audio-visual transformer for multimodal fake media detection.	FakeAVCeleb	A unified transformer that processes both audio and visual tokens simultaneously (Dataset generation)	Better performance by modeling cross-modal interactions from the start. ~65% AUC benchmark	Reduce the high computational cost of audio-visual transformers.

Analysis of Survey Paper

The most successful and common approach was focusing on detecting inconsistencies between frames and a multimodal approach. The work by Afchar et al. [8] emphasized compact

CNNs for efficiency (achieved 89-98% accuracy on facial forgeries), influenced later by temporal models [2], [4]. Multimodal shifts in the works by Zhou et al. [9] and Kniaz et al.[10] address audio-video integration, boosting generalization by 65-95% but it did highlight challenges like bias and compression, which are common occurrences when training a model.

More recent papers addressed by the works [1 - 7] build on these by targeting Sora-like generators (AI Video generators), with transformers and few-shot learning closing the gaps in detection. The most common strengths for all of the related works are via fusion or multimodality and a frame-based detection strategy is not only a viable approach but highly promising, as demonstrated by the work in [21] However, it is explicitly a preliminary study conducted on a very small and limited dataset, only 300 annotated frames from 15 Sora videos, compared to other approaches. The need for a multimodal approach and a new dataset focused on newer, and more capable AI video generators, that are able to generate both audio and visual synthetic data.

Chapter 3

METHODOLOGY

The section covers the implementation of a multimodal detection system for synthetic content generated primarily by Sora 2 and Veo 3 generated content. These tools are able to generate both audio and visual data. This study is a quantitative experimental design to assess the performance of the proposed model. It will try to determine the optimal deep learning methods and to measure their classification accuracy compared to existing techniques. The proposed framework is a hybrid/multimodal model on a selected dataset of real and AI-generated audio

and visual content, primarily Sora 2 and Veo 3. The methodology is loosely based on the works done by Gandhi et al [27] and Rodriguez et al. [32] with its approach on a multimodal framework on deepfake detection and AI image detection.

The dataset is separated into 2 groups;

a) Synthetic (AI Generated) Dataset: Created using state-of-the-art generative tools such as Sora 2 and Veo 3, which generated both synthetic visual and auditory content. The model that will be developed would try to identify the artifacts and signatures of these tools.
b) Authentic Dataset: This will serve as the negative class representing real world content. Sourced from DFDC Dataset (Video) and Fake-or-Real (FoR) dataset (Audio). This will ensure a high-quality "real" dataset to be compared to the synthetic dataset.

Data Gathering

The dataset will mostly compose of Sora 2 and Veo 3 synthetic user-generated content, which are publicly available and downloadable, and authentic data from the real world. The baseline would be at first, 50 Sora 2, 50 Veo 3 videos and 100 real world videos from the DFDC Dataset which would then be scaled up depending on the results of the mode and how well it performs.

The dataset for the audio would be sourced from both Sora 2 and Veo 3, which would then be compared with 100 2-4 seconds in duration human speech sourced from Fake-or-Real (FoR) dataset, which is publicly available and readily downloadable.

Data Preprocessing and Feature Extraction

The collected synthetic video stream would be sliced into image frames, with the use of Python scripts and each video will be processed to a standard format, which is 1080p, as both Veo 3 and Sora 2 supports it. Both Veo 3 and Sora 2 produce 10-second length, 24 frame per second videos, totalling over 240 frames individually for one video. Using PRNU (Photo Response Non-Uniformity) extraction, comes from the different light sensitivity of the different pixels (elementary sensors). This is an unavoidable characteristic due to manufacturing imperfections, and it is present on all image sensor chips [31].

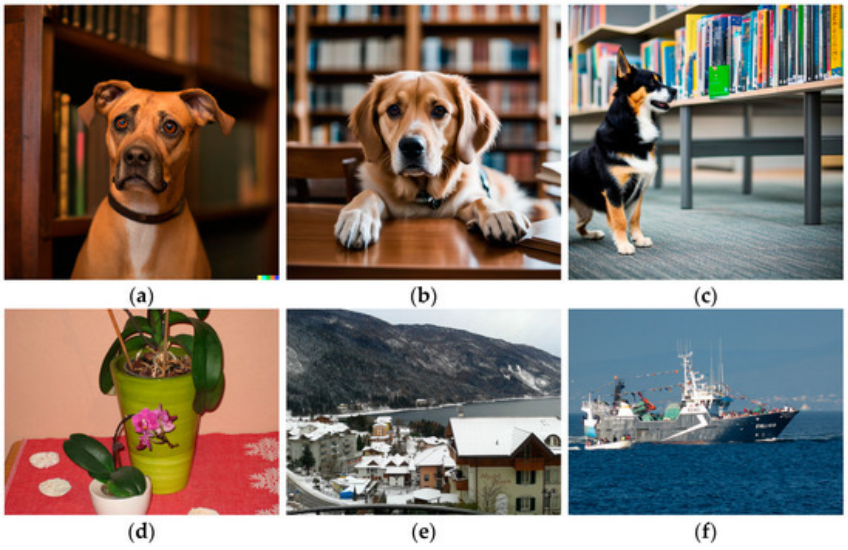


Figure 2a. (a) DALL E 2 image, (b) Stable Diffusion image, and (c) OpenArt image. (d–f) Real photos [30].

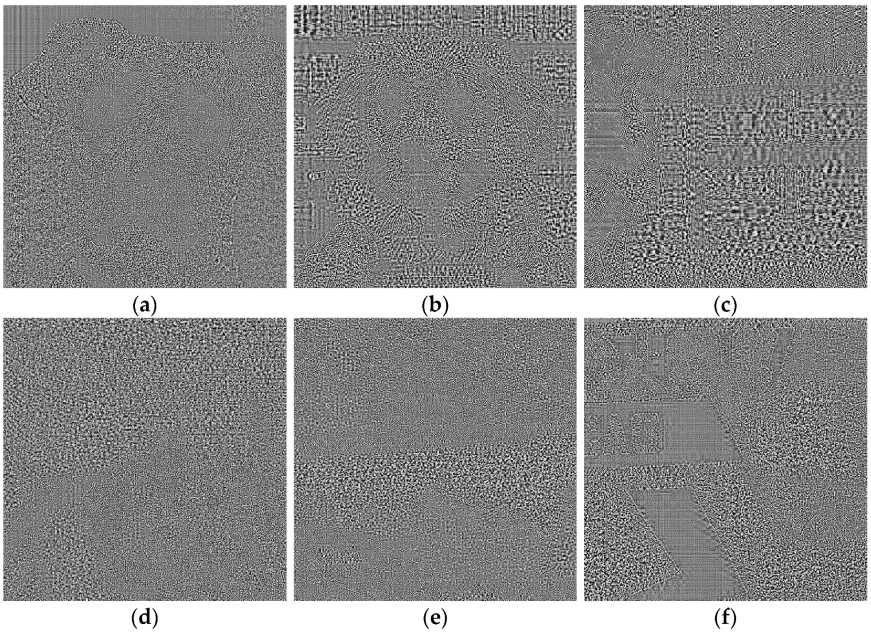


Figure 2b. (a–c), PRNU patterns computed for AI images of Figure 2a. (d–f) are examples of PRNU patterns for real images [30].

The audio track will be extracted and converted into Mel-spectrograms [30] using the Librosa library. Mel-spectograms provide a time-frequency representation of audio signals with perceptually relevant amplitude and frequency representations [27].

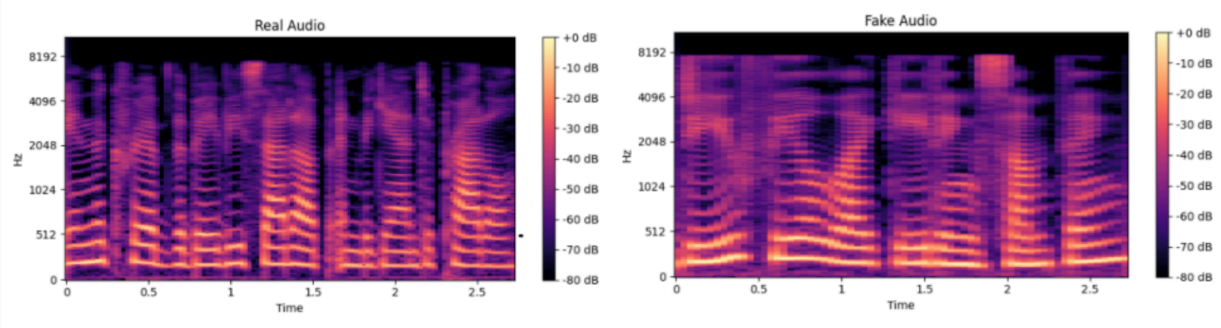


Figure 3. Mel-spectrograms comparing real (left) and deepfake (right) audio signals [27].

Model Architecture

A hybrid deep learning model will be implemented using Python with the PyTorch framework. The visual stream will be processed by a pre-trained Convolutional Neural Network (CNN) or EfficientNetV2, since it is known for its high efficiency and accuracy in feature extraction. A work done in [30] used CNN with the following structure shown below:

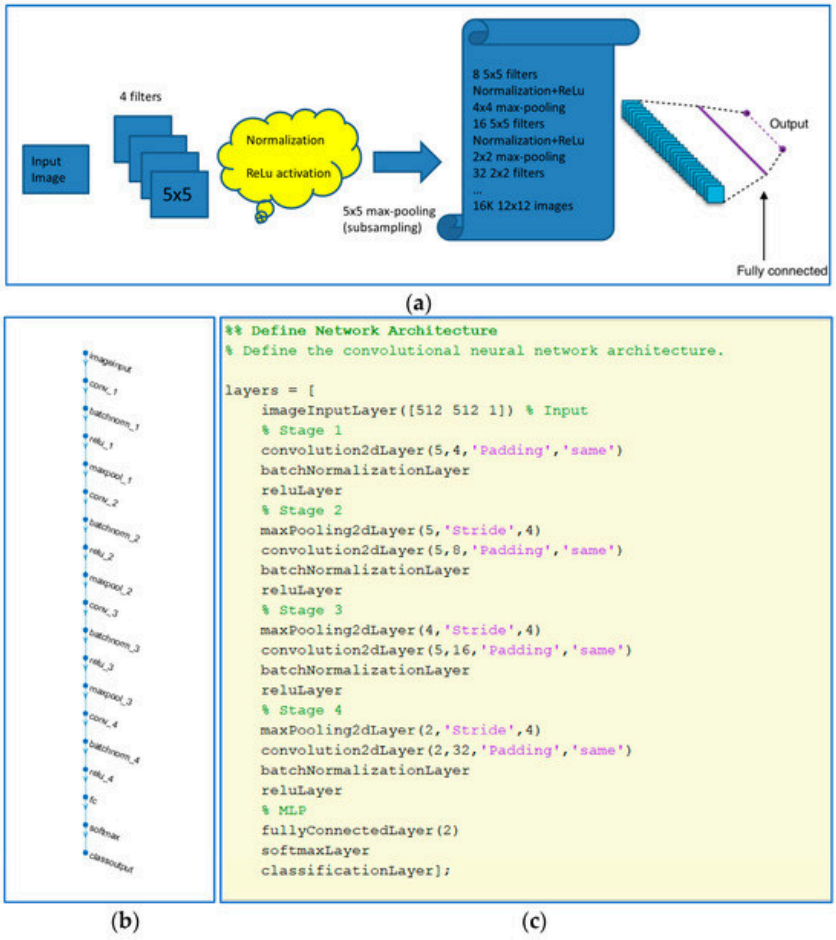


Figure 4. (a) CNN structure used. (b) Layers diagram. (c) Matlab code used to define net structure [30].

For the audio stream, Random Forest and XGBoost can be used to classify labeled audio samples due to their strong performance in complex data processing. CNNs will also be used for

classifying mel spectrograms because CNNs excel at identifying and learning spatial hierarchies in image-like data [27].

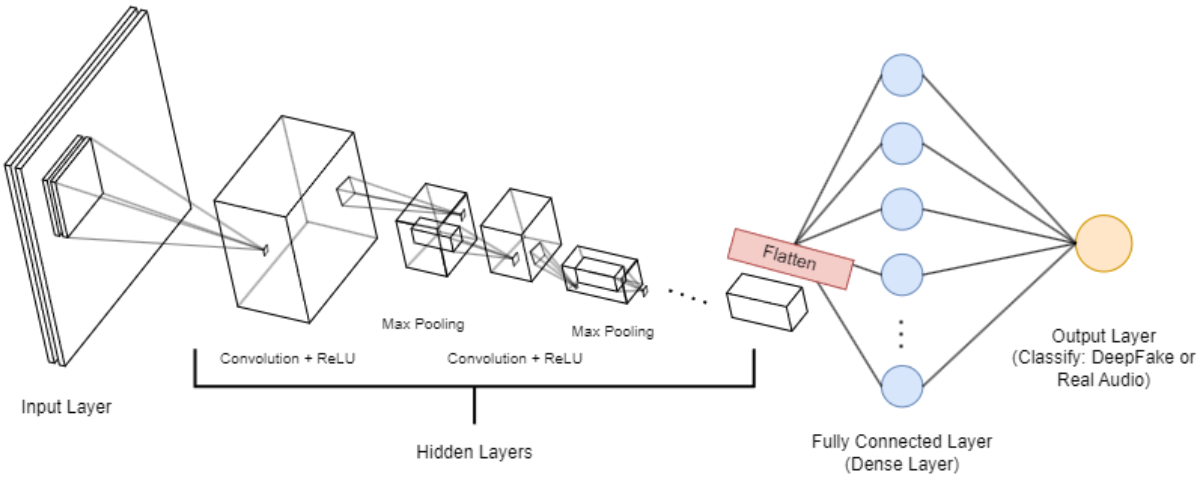


Figure 5. Architecture of the Convolutional Neural Network (CNN) used for classifying mel-spectrograms, highlighting layers such as Conv2D and MaxPooling2D for feature extraction and classification [27].

Training and Evaluation

The model would most likely be trained on the training dataset using the Adam optimizer and binary cross-entropy loss function. This is still a work in progress, and other techniques could be better suited in the process. Transfer learning will be employed, leveraging the pre-trained weights of the CNNs to accelerate training and improve performance.

The final trained model will be evaluated on unseen testing dataset. It will be measured with Accuracy, Precision, Recall and F1-Score. The instruments will be Python, PyTorch, OpenCV, Librosa, Scikit-learn, FFmpeg, Mel-Spectogram, and PRNU.

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